

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming (Mock Interview)

Course Name	Python Programming
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Topic	CO3-AT3-MOCK INTERVIEW
Question	Q3 — Trace the Output: Dictionary Operations (10 Marks)

Q3. [10 Marks] Trace the output:

```
d = {'a': 1, 'b': 2, 'c': 3}
d.update({'b': 20, 'd': 40})
print(d)
del d['a']
print(d.pop('z', 'Not found'))
print(d)
```

Answer

Line-by-line trace

```
d = {'a': 1, 'b': 2, 'c': 3}
```

Creates a dictionary d with three key-value pairs: 'a':1, 'b':2, 'c':3.

```
d.update({'b': 20, 'd': 40})
```

The update() method merges the given dictionary into d. Since key 'b' already exists, its value is overwritten from 2 to 20. Since key 'd' does not exist, it is added as a new pair. d is now {'a': 1, 'b': 20, 'c': 3, 'd': 40}.

```
print(d)
```

Prints the dictionary as it stands after the update — {'a': 1, 'b': 20, 'c': 3, 'd': 40}. Dictionaries in Python 3.7+ preserve insertion order, so keys appear in the order they were first added.

```
del d['a']
```

The del statement removes the key 'a' (and its value 1) from the dictionary entirely. d is now {'b': 20, 'c': 3, 'd': 40}.

```
print(d.pop('z', 'Not found'))
```

pop() attempts to remove key 'z' and return its value. Since 'z' does not exist in d, pop() returns the supplied default, 'Not found', instead of raising a KeyError. The dictionary itself is left unchanged because no matching key was found to remove.

```
print(d)
```

Prints the final state of the dictionary, which is unchanged from the previous step: {'b': 20, 'c': 3, 'd': 40}.

Key concepts used

- **dict.update(other)**: merges another dictionary (or iterable of key-value pairs) into the calling dictionary — existing keys are overwritten with the new value, and new keys are inserted.
- **del dict[key]**: deletes a key and its associated value from the dictionary; raises a `KeyError` if the key does not exist.
- **dict.pop(key, default)**: removes and returns the value for key if it exists; if the key is absent, it safely returns the given default instead of raising an error, unlike `dict.pop(key)` used without a default.

Verified Program

```
"""
CO3-AT3 - Mock Interview
Q3: Trace the Output - Dictionary Operations [10 Marks]
-----
Verifies the step-by-step trace of update(), del, and pop()
on a Python dictionary.
"""

d = {'a': 1, 'b': 2, 'c': 3}
d.update({'b': 20, 'd': 40})
print(d)

del d['a']
print(d.pop('z', 'Not found'))
print(d)
```

Actual Output

```
{'a': 1, 'b': 20, 'c': 3, 'd': 40}
Not found
{'b': 20, 'c': 3, 'd': 40}
```

Conclusion

Tracing this snippet shows how a Python dictionary evolves as different mutating methods are applied to it. `update()` merges in new key-value pairs and overwrites existing keys in place; `del` removes a key outright and raises an error if it is missing; and `pop()` with a default value offers a safe way to remove a key that fails gracefully by returning a fallback string, 'Not found', instead of crashing the program when the key does not exist. The final dictionary, `{'b': 20, 'c': 3, 'd': 40}`, reflects the cumulative effect of the update and the deletion, while the `pop()` call on a non-existent key leaves the dictionary unchanged.